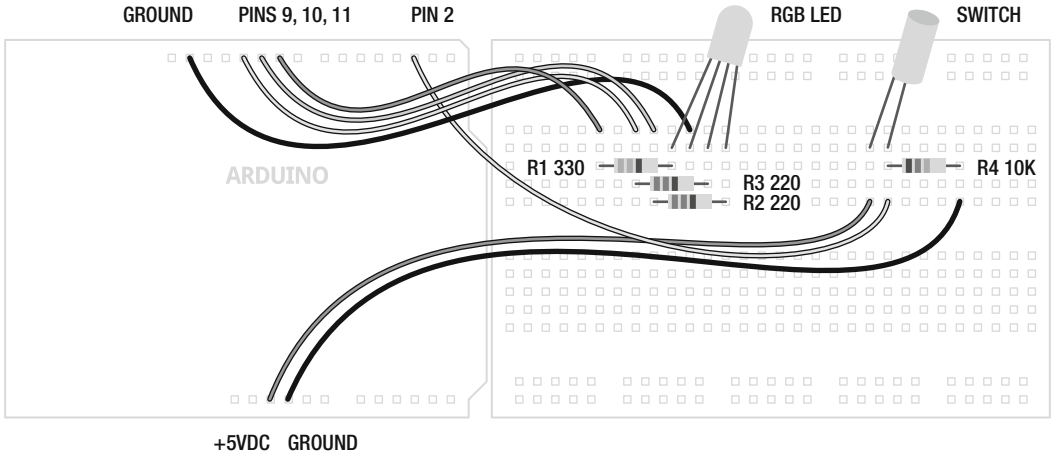


Figure 4-2. *Tilt Blink illustration*

Uploading the Source Code

In our last project, we managed to cycle through seven different colors with our LED in a fairly simple and straightforward manner. And that's generally a good thing, except when you start to try to do two or more things at once like cycling through different colors all the while waiting for a tilt switch to be triggered. In the last example, it would take 8 seconds to make it through all of the colors once. In that amount of time we could have easily missed a switch being triggered because our code was not smart enough, or fast enough, to do two things at once. In our source code for this project, shown in Listing 4-1, we have attempted to solve this by speeding things up a bit, reducing our time between readings from 8 seconds to a half of a second. We've also compartmentalized our code a bit more, introducing many of the concepts that I will explain further in this chapter.

Listing 4-1. *Tilt Blink Source Code*

```
const int rgb[] = {9,10,11};
const int time = 250;
const int switchPin = 2;

void setup() {
  for (int i=0; i<3; i++) pinMode(rgb[i], OUTPUT);
  pinMode(switchPin, INPUT);
}

void loop() {
  int newPin = 0;
  int oldPin = 0;
```